MANUSCRIPT REVIEW REPORT

Assessing the Carbon Footprint of Virtual Meetings: A Quantitative Analysis of Camera Usage

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ABOUT THIS REPORT

This report was written by checkmymanuscript, an AI-powered manuscript review tool designed to help researchers and authors avoid common errors before submission. Used by researchers from leading universities including Stanford, Johns Hopkins, New York University, and more, our agentic workflow identifies potential issues across multiple dimensions including formatting consistency, citation accuracy, language quality, and structural completeness. The goal is to catch common errors and provide suggestions, helping you present your best work to journals and reviewers.

OVERALL SUMMARY

The manuscript requires substantial revisions, particularly concerning the structural organization of content, the completeness and accuracy of references, and language precision. Addressing these critical areas will significantly enhance the manuscript's clarity, adherence to academic standards, and overall readiness for submission.

CRITICAL ISSUES	MAJOR ISSUES	MINOR ISSUES
0	24	2

LANGUAGE QUALITY

Overall Language Score: A-

The manuscript presents a clear research question and methodology. Language quality is generally good, though occasional subject-verb agreement errors and minor inconsistencies in numerical formatting slightly detract from overall clarity and publication readiness.

Category Assessments

Grammar and Syntax

C+

Frequent subject-verb agreement errors, particularly with 'data' and citations, impact grammatical correctness. Punctuation and sentence structure are otherwise mostly sound.

Clarity and Precision

B+

The core research question and findings are clearly communicated. Phrasing is generally precise, although some sentences could be more direct.

Conciseness

A-

The writing is largely concise, avoiding unnecessary jargon or overly lengthy explanations. Redundancy is minimal throughout the analyzed text.

Academic Tone

A

The tone is appropriately formal and scholarly, suitable for an academic audience. Objective language is used effectively to present research.

Consistency

B

Minor inconsistencies noted in numerical formatting (e.g., '6 000') and capitalization ('Netflix'). Terminology and verb tense usage appear consistent.

Readability and Flow

B+

Logical structure and clear transitions aid readability. Sentence variety is adequate, though some areas could benefit from smoother phrasing.

Key Strengths

- Clear articulation of the research problem and methodology.
- Appropriate and consistent academic tone maintained throughout.
- Logical organization and flow support reader comprehension.

Areas for Improvement

- Recurring subject-verb agreement errors, especially with collective nouns and citations.

- Minor inconsistencies in numerical formatting require attention.
- Occasional awkward phrasing or slightly imprecise sentence constructions.

SECTION REVIEW SUMMARY

SECTION	STATUS	ISSUES
Language Evaluates tone, grammar, and overall readability.	✗	8
Abstract Ensures the abstract summarises scope, approach, and key findings.	✗	4
Referencing Checks citation completeness and reference formatting.	✗	4
Title Page Checks that title page metadata and authorship details are complete.	✗	3
Structure Verifies the manuscript includes the required structural components.	✗	3
Figures and Tables Assesses presence, captions, and referencing of visuals.	✗	2
Title Confirms the title is clear and aligned with the manuscript.	✗	1
Funding Verifies acknowledgements and funding disclosures are included.	✗	1
Main Headings Reviews heading hierarchy and formatting consistency.	✓	
Acronyms Confirms acronyms are introduced and used consistently.	✓	

TITLE RECOMMENDATIONS

1 issue: 1 Major

The revised title is specific and accurately reflects the manuscript's key findings and focus on mobile networks.

Location: Title

Major

Original: Assessing the Carbon Footprint of Virtual Meetings: A Quantitative Analysis of Camera Usage

Suggestion: Camera Off in Video Calls Halves Mobile Data Consumption and Carbon Footprint

Explanation: The original title is too general. The revised title is more specific, incorporates key findings (halving data consumption and carbon footprint), and highlights the focus on mobile networks as indicated in the abstract and section summaries.

TITLE PAGE RECOMMENDATIONS

3 issues: 3 Major

Keywords need refinement for specificity. Institutional affiliations and email addresses require correction for academic professionalism and completeness.

Location: Keywords

Major

```
Suggestion: Keywords: carbon footprint; virtual meetings; video conferencing; data consumption; camera usage; environmental impact; mobile networks
```

Explanation: The original keywords ['data consumption', 'video calls', 'carbon footprint', 'mobile networks'] are relevant but could be more specific and comprehensive. 'Video calls' is generic; 'video conferencing' or 'virtual meetings' would be more appropriate. 'Camera usage' is a key factor investigated and should be included. 'Environmental impact' is a broader concept that the carbon footprint relates to and is good to include. Adding 'virtual meetings' directly from the title would also enhance discoverability. The suggested list improves specificity and covers the core aspects of the paper more thoroughly, as indicated by the title and abstract.

Location: Authors

Major

```
Original: Independent Researcher\Paris, France
```

Explanation: The institutional affiliation for Félix Mortas is incomplete. Please add the department and ensure the city and country are clearly separated. For example: 'Department, Institution, City, Country'.

Location: Authors

Major

```
Original: felix.mortas@hotmail.fr
```

Explanation: A personal email address (@hotmail.fr) is used. It is recommended to use an institutional email address for academic publications to ensure professional correspondence.

ABSTRACT RECOMMENDATIONS

4 issues: 2 Major 2 Minor

Abstract phrasing needs improvement for grammatical accuracy, directness, and precision in describing the study's purpose and methodology.

Location: Abstract

Major

Original: The study is used to quantify the real reduction in environmental impact claimed in several articles when people choose to turn off their camera during meetings.

Suggestion: The study **quantifies** the reduction in environmental impact **achieved by turning off cameras during meetings, challenging some prevalent claims.**

Explanation: The original phrasing is awkward and less direct. The suggested text clearly states the study's purpose in relation to existing claims.

Location: Abstract

Major

Original: This paper analyzes the carbon emissions related to data consumption during video calls, focusing on the impact of having the camera on versus off.

Suggestion: This paper **quantifies** the carbon **footprint associated with** data consumption during video calls, **specifically analyzing** the impact of camera **usage (on** versus **off).**

Explanation: The term 'analyzes' is vague; 'quantifies' is more precise and aligns with the paper's quantitative methodology and results.

Location: Abstract

Minor

Original: The findings indicate that turning the camera off can halve data consumption therefore carbon emissions, particularly on mobile networks, and conclude with recommendations to optimize data usage and reduce environmental impact during calls.

Suggestion: The findings indicate that **deactivating** the camera can halve data consumption **and associated** carbon emissions, **especially** on mobile **networks. The paper concludes** with recommendations **for optimizing** data usage and **reducing the** environmental impact **of** calls.

Explanation: The phrase 'therefore carbon emissions' is grammatically awkward. Separating the conclusion about recommendations into a new sentence improves readability and clarity.

Location: Abstract

Minor

Original: The experiment was carried out using a 4G connection via a cell phone to understand the varying data transfer associated with videos.

Suggestion: The **study utilized** a 4G **mobile connection** to **measure** data transfer **during video calls.**

Explanation: The original phrasing is redundant. The revised text is more concise while maintaining essential methodological information.

LANGUAGE RECOMMENDATIONS

8 issues: 8 Major

Numerous grammatical errors, including subject-verb disagreements and incorrect numerical/proper noun formatting, need correction for clarity and academic rigor.

Location: Data Overview and Preliminary Analysis

Major

Original: electricity consumption in kWh per GB is 2.6 to 6 000 times higher

Suggestion: electricity consumption in kWh per GB is 2.6 to **6000** times higher

Explanation: Corrected numerical formatting: '6 000' should be '6000' for clarity and consistency.

Location: Data Overview and Preliminary Analysis

Major

Original: \cite{EuropeanCommission2020} report an energy consumption

Suggestion: \cite{EuropeanCommission2020} **reports** an energy consumption

Explanation: Subject-verb disagreement: The citation \cite{EuropeanCommission2020} is treated as a singular entity, so the verb should be 'reports' not 'report'.

Location: Data Overview and Preliminary Analysis

Major

Original: \cite{Golard2020} present data for Belgium

Suggestion: \cite{Golard2020} **presents** data for Belgium

Explanation: Subject-verb disagreement: The citation \cite{Golard2020} is treated as a singular entity, so the verb should be 'presents' not 'present'.

Location: Interpretation

Major

Original: For video calls, having the camera on results in emissions ranging from 0.17 to 0.36 grams of CO2e per minute, which translates to 10.2 to 21.6 grams over an hour.

Suggestion: For video calls, having the camera on results in emissions ranging from 0.17 to 0.36 grams of CO2e per minute, which **translate** to 10.2 to 21.6 grams over an hour.

Explanation: Fixed subject-verb disagreement: 'which translates' to 'which translate' because 'which' refers to the plural 'emissions ranging'.

Location: Interpretation

Major

Original: Based on \ref{tab:co2emissions}, a one-hour video call with a camera on is equivalent to about a 1-minute hot shower or a 30-minutes netflix session.

Suggestion: Based on \ref{tab:co2emissions}, a one-hour video call with a camera on is equivalent to about a 1-minute hot shower or a **30-minute Netflix** session.

Explanation: Corrected grammatical error: '30-minutes' to '30-minute' to function as a proper adjective, and capitalized the proper noun 'Netflix'.

Location: Introduction

Major

```
Original: Although data has been found for electricity consumption
per amount of data transferred on fixed networks ranging from
0.00002 to 0.1 kWh/GB, we're going to retain ARCEP's assumption
that the amount of electricity consumed varies little according to
the amount of data transferred for the continuation of the study,
while bearing in mind that it still has a small impact.
```

```
Suggestion: Although data have been found for electricity
consumption per amount of data transferred on fixed networks
ranging from 0.00002 to 0.1 kWh/GB, we're going to retain ARCEP's
assumption that the amount of electricity consumed varies little
according to the amount of data transferred for the continuation of
the study, while bearing in mind that it still has a small impact.
```

Explanation: Fixed subject-verb agreement: 'data has been found' to 'data have been found'. 'Data' is a plural noun, and thus requires a plural verb.

Location: Introduction

Major

```
Original: Data for Europe, provided by
\cite{EuropeanCommission2020}, is showing that fixed networks
consume 0.03 kWh/GB.
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```
Suggestion: Data for Europe, provided by
\cite{EuropeanCommission2020}, are showing that fixed networks
consume 0.03 kWh/GB.
```

Explanation: Corrected subject-verb agreement: 'Data... is showing' to 'Data... are showing'. 'Data' is a plural noun, and thus requires a plural verb.

Location: Introduction

Major

```
Original: Data for the USA from \cite{Thundersaid2024} shows that
fixed fiber networks consume 0.04 kWh/GB.
```

```
Suggestion: Data for the USA from \cite{Thundersaid2024} show that
fixed fiber networks consume 0.04 kWh/GB.
```

Explanation: Corrected subject-verb agreement: 'Data... shows' to 'Data... show'. 'Data' is a plural noun, and thus requires a plural verb.

STRUCTURE RECOMMENDATIONS

3 issues: 3 Major

Content organization requires significant adjustment. Analysis and interpretation should be moved to Results and Discussion, respectively, and recommendations integrated.

Location: Data Overview and Preliminary Analysis

Major

```
Original: Data Overview and Preliminary Analysis
```

Explanation: The content of this section, which details data analysis and carbon emission calculations, should be moved to a Results section. Results should present findings, not preliminary analysis.

Location: Interpretation

Major

```
Original: Interpretation
```

Explanation: The content of this section, which interprets preliminary data and compares emissions, should be integrated into the Discussion section. Interpretation is a core component of the Discussion.

Location: Recommendations

Major

```
Original: Recommendations
```

Explanation: The 'Recommendations' section is not a standard top-level section in academic papers. Integrate its content into the Discussion or Conclusion section to align with standard academic conventions.

FIGURES AND TABLES RECOMMENDATIONS

2 issues: 2 Major

Figures and tables need enhanced clarity through the inclusion of units of measurement and specific examples like network types.

Location: Interpretation

Major

Original: CO2 emissions of different activities.

Suggestion: CO2 emissions of different **activities (kg CO2eq).**

Explanation: Enhanced clarity by including units of measurement (kg CO2eq) to provide essential quantitative context for the reported CO2 emissions.

Location: Introduction

Major

Original: Energy consumption of different networks.

Suggestion: Energy consumption of different **networks (e.g., 4G).**

Explanation: Enhanced clarity by specifying an example network (e.g., 4G) that is supported by the document context.

REFERENCING RECOMMENDATIONS

4 issues: 4 Major

Significant issues exist with missing references, incomplete author information, and a lack of source diversity, requiring thorough verification and revision.

Location: Introduction

Major

```
Original: The Information and Communication Technology (ICT) sector
is responsible for approximately 2 to 4\% of global greenhouse gas
(GHG) emissions \cite{Freitag2021}.
```

Explanation: The reference 'Freitag2021' is cited in the text but not found in the bibliography.

Location: Introduction

Major

```
Original: With the advent of generative AI, reputable institutions
project a significant increase in these emissions in the near
future \cite{Iea2025}.
```

Explanation: The reference 'Iea2025' is cited in the text but not found in the bibliography.

Location: Introduction

Major

```
Original: The most popular and serious study on this subject is
highlighting a reduction in 96\% environmental impact by turning
off cameras during virtual meetings \cite{Obringer2021}.
```

```
Suggestion: The most popular and serious study on this subject
highlights a reduction in 96\% environmental impact by turning off
cameras during virtual meetings \cite{Obringer2021}.
```

Explanation: The citation 'Obringer2021' is missing from the bibliography. Additionally, the subject-verb agreement needs correction: change 'study is highlighting' to 'study highlights'.

Location: Bibliography

Major

Explanation: Several references in the bibliography are missing author information. Please verify and add the authors for 'Prysmian2021', 'Thundersaid2024', 'Pihkola2018', 'EuropeanCommission2020', 'Traficom2022', 'Sfr', 'Hp', 'Windows10', 'Teams', 'Nokia2025', 'Golard2020', 'Arcep2019', 'Baseempreinte', 'Carbonfootprintmodelingtool', and 'Sherman2013'. Additionally, the bibliography demonstrates venue bias and a lack of source diversity, as all citations originate from a single, unspecified venue. Revision is needed to include actual research references and assess timeliness.

FUNDING RECOMMENDATIONS

1 issue: 1 Major

A funding statement is absent and must be added to acknowledge financial support, following standard academic conventions.

Location: Funding Statement

Major

Explanation: No funding statement was found. A funding statement briefly acknowledges the financial support behind a research project. It typically mentions the funding agency, the grant number, and sometimes the program name. It's usually placed in the acknowledgments or before the references. For example: "This work was supported by the European Research Council (ERC) under the European Union's Horizon 2020 programme (Grant agreement No. 758892)." or "The research was funded by the National Institutes of Health (NIH) under Grant R01 GM123456."